Validation and Calibration of the Remote Consolidated Approach of Reporting on Food Security (rCARI) Methodology

Preliminary results from Myanmar, Ecuador, and Central African Republic

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November 2, 2023

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1 Introduction

The overall goal of this study was to assess the validity of the rCARI module(s) against conventional CARI modules for measuring food security within the same population. The study will compare the indicators measured through phone surveys with conventional face-to-face survey indicators, considering the latter as the so-called "gold standard." The following food security indicators were compared (FCS, LCS, rCSI, FES, income sources, and income change). The ECMEN indicator will not be tested in this study, as it requires a standard expenditure module, which cannot be collected over the phone. A shortened version of the standard expenditure module will be used to test the possibility of collecting FES data over the phone.

Studying the validity of the phone modality in collecting food security indicators is a key step in judging the appropriateness of using the rCARI module to estimate food security prevalence. If needed, the validation will be followed by calibration to ensure that the modality of data collection has minimal influence on the estimate (rCARI).

However, several other factors should be considered while reading estimates of any indicator (whether it is food security or not) when it comes from a nationally representative phone survey.

- Mobile phone ownership (mobile phone penetration rate) excluded the most vulnerable population from the sample. In some countries where WFP operates, the penetration rate is as low as 30%, whereas in others, it exceeds 100%.
- Mobile phone network coverage: Even in countries with high penetration rates, network coverage could be a problem, especially when electricity cuts are frequent and local infrastructure, such as communication masts, are not evenly distributed (World Bank, 2020).
- Response rate: It is well documented that the response rate in phone surveys is usually lower than in face-to-face surveys (L'Engle et al., 2018; World Bank, 2020).

These factors usually lead to a skewed sample; nevertheless, there are several weighting techniques to adjust the results of phone surveys based on the demographics of the most recent face-to-face surveys (Nagpal et al. 2021; Glazerman et al. 2020; Valliant et al. 2013; Elkasabi, 2015).

This study does not cover weighting techniques to adjust for the sampling frame and response rate issued to produce a national estimate. This has been covered in the literature, and several studies have been conducted to control for these biases.

2 Study design

The study was implemented in a randomized controlled setup. Food security data were collected from two statistically comparable groups of households within the same sample. Stand-alone standardized remote and face-to-face questionnaires will be administered to sampled households randomized to receive either modalities-remote or face-to-face. The study will be implemented across six countries: two in Africa, two Latin America, one in Southeast Asia, and one in the Middle East.

2.1 Selection of study areas and participants

The selection of study villages proceeded as described in (figure 1).Listing of potential households/respondents to form the sampling frame after taking their informed consent (a minimum of 100 households per village/lowest admin area/camp, i.e., 2000hh in total will be listed through a random walk approach). For purposes of statistical comparison, demographic information will collected from all households contacted during the listing phase. Households that agreed to be listed for the study are randomly assigned into one of two modalities (groups) in a pure random assignment; 1000 households receiving the remote/phone interview (called the treatment group) and 1000 households assigned to receive the face-to-face interview (called the control group). The study will aim to achieve a minimum of 600 completed interviews for each modality from the originally assigned groups (i.e., there will be no replacement for non responses).

2.2 Implementation

Fieldwork was originally planned to take place over a 3 week period (covering 1 week for enumerator training, adjusting questionnaire and 2 weeks for actual household interviews) in each pilot country. However, because of unforeseen challenges as will be discussed later, data collection was extended by between 2 to 3 weeks depending on the country.

3 Sample Completion

Table 1 Shows the total sample and households interviews completed by country and modality in the initial 3 pilot countries of the study. Generally, the figures in Table 1 show that we have lower response rate across countries for the remote modality.

Table 1: Survey Completion by Country

	Country			
	Myanmar	Ecuador	CAR	Total
N	1,665 (27.2%)	2,372 (38.8%)	2,080 (34.0%)	6,117 (100.0%)
Survey Progress				
Not Completed	572 (34.4%)	956 (40.3%)	835 (40.1%)	2,363 (38.6%)
Completed	1,093 (65.6%)	1,416 (59.7%)	1,245 (59.9%)	3,754 (61.4%)
Survey Modality				
Face-to-Face	610 (55.8%)	760 (53.7%)	969 (77.8%)	2,339 (62.3%)
Remote	483 (44.2%)	656 (46.3%)	276 (22.2%)	1,415 (37.7%)

4 Primary Results: Impact of Survey Modality

In this section, we present the main results of the study aggregated by the food security indicator. Here, the main objective is to compare the different CARI variations aggregated either using remote or face-to-face modules. CARI aggregated using remote modules is referred to us rCARI (remote CARIR). Depending on the indicators used, we have CARI-FES and CARI-Inc which is CARI aggregated using the standard FES (food expenditure share) and income and income change variables respectivey. For the purposes of this interim analysis and report, we restrict the CAR sample to only respondents that owned phones (the complete sample comprising respondents that received phones is available upon request).

4.1 CARI vs. rCARI

There are four versions of overall indicators we used in the validation study:

- CARI-FES: Standard CARI using FES module from F2F surveys
- rCARI-FES: CARI using FES module from Remote surveys
- CARI-Inc: CARI using Income & income change from F2F surveys
- rCARI-Inc: Standard rCARI using Income & income change module from Remote surveys

From Table 2, we have three different CARI aggregation methods to test against the standard/gold-standard CARI. We will further compare the results of rCARI aggregated using FES with rCARI using the income and income change variables.

Table 2: Lists the different aggregation methods and indicators used for each methodology

	Aggregation Methodology			
Indicator	**CARI (F2F)**	**CARI (F2F)**	**rCARI (Remote)**	**rCARI (Remote)**
FCS	Yes	Yes	Yes	Yes
FES	Yes	**No**	Yes	No
LCS	Yes	Yes	Yes	Yes
rCSI	Yes	Yes	Yes	Yes
Income source & income change	**No**	Yes	**No**	**Yes**

Table 3 presents the first comparison of rCARI (using the income and income change variable) with the standard CARI across the 3 pilot countries. The results from this table show two key findings. First, except for CAR, the calculated food insecurity category (moderately food insecure plus severely food insecure categories) did not differ significantly between the two methods. Second, although the results were not statistically different between the two aggregation methods, rCARI is always the conservative method (reporting lower food insecure numbers) relative to stanadard CARI.

Table 4 shows the results of rCARI aggregated using the FES module collected over the phone,

Table 3: Impact of Survey Modality: Standard CARI-FES vs. Standard rCARI-Inc

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
CARI vs. rCARI: Food Secure	0.08	0.07	-0.01	0.01	0.00	-0.01	0.01	0.01	-0.00
	(0.27)	(0.26)	(0.67)	(0.11)	(0.07)	(0.08)	(0.08)	(0.08)	(0.96)
CARI vs. rCARI: Marginally Food Secure	0.44	0.41	-0.03	0.23	0.27	0.03	0.31	0.52	0.21***
	(0.50)	(0.49)	(0.42)	(0.42)	(0.44)	(0.13)	(0.46)	(0.50)	(0.00)
CARI vs. rCARI: Moderately Food Insecure	0.44	0.49	0.05	0.59	0.62	0.03	0.60	0.45	-0.15**
	(0.50)	(0.50)	(0.23)	(0.49)	(0.48)	(0.22)	(0.49)	(0.50)	(0.00)
CARI vs. rCARI: Severely Food Insecure	0.03	0.03	-0.01	0.16	0.11	-0.06**	0.08	0.02	-0.06**
	(0.18)	(0.16)	(0.59)	(0.37)	(0.31)	(0.00)	(0.27)	(0.14)	(0.00)
CARI vs. rCARI: Food Insecure	0.47	0.51	0.04	0.76	0.73	-0.03	0.68	0.47	-0.21***
	(0.50)	(0.50)	(0.31)	(0.43)	(0.44)	(0.26)	(0.47)	(0.50)	(0.00)
Observations	610	232	842	760	656	1,416	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

Table 4: Impact of Survey Modality: CARI-FES vs. rCARI-FES

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
CARI using FES: Food Secure	0.08	0.03	-0.06***	0.01	0.00	-0.01**	0.01	0.01	-0.00
	(0.27)	(0.16)	(0.00)	(0.11)	(0.04)	(0.01)	(0.08)	(0.08)	(0.96)
CARI using FES: Marginally Food Secure	0.44	0.34	-0.10**	0.23	0.07	-0.16***	0.31	0.29	-0.02
	(0.50)	(0.48)	(0.01)	(0.42)	(0.26)	(0.00)	(0.46)	(0.45)	(0.61)
CARI using FES: Moderately Food Insecure	0.44	0.57	0.13***	0.59	0.69	0.10***	0.60	0.58	-0.02
	(0.50)	(0.50)	(0.00)	(0.49)	(0.46)	(0.00)	(0.49)	(0.50)	(0.65)
CARI using FES: Severely Food Insecure	0.03	0.06	0.02	0.16	0.23	0.07**	0.08	0.13	0.05
	(0.18)	(0.23)	(0.17)	(0.37)	(0.42)	(0.00)	(0.27)	(0.33)	(0.14)
CARI using FES: Food Insecure	0.47	0.63	0.16***	0.76	0.92	0.17***	0.68	0.71	0.02
	(0.50)	(0.48)	(0.00)	(0.43)	(0.27)	(0.00)	(0.47)	(0.46)	(0.61)
Observations	610	232	842	760	656	1,416	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

which is a relatively shorter version of the standard FES usually collected in F2F surveys. This version collects the household's aggregate expenditure without going through individual items. The results show that rCARI using FES consistently over-states the number of food insecure across all countries compared to the standard CARI by between 2 and 17 percentage points. This difference was statistically significant in Myanmar and Ecuador.

Table 5: Impact of Survey Modality: CARI-Inc vs. rCARI-Inc

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
rCARI using Income: Food Secure	0.16	0.07	-0.09***	0.03	0.00	-0.02***	0.02	0.01	-0.01
	(0.37)	(0.26)	(0.00)	(0.17)	(0.07)	(0.00)	(0.14)	(0.08)	(0.18)
rCARI using Income: Marginally Food Secure	0.43	0.41	-0.02	0.33	0.27	-0.07**	0.51	0.52	0.01
	(0.50)	(0.49)	(0.68)	(0.47)	(0.44)	(0.01)	(0.50)	(0.50)	(0.86)
rCARI using Income: Moderately Food Insecure	0.39	0.49	0.10**	0.53	0.62	0.09***	0.45	0.45	0.01
	(0.49)	(0.50)	(0.01)	(0.50)	(0.48)	(0.00)	(0.50)	(0.50)	(0.89)
rCARI using Income: Severely Food Insecure	0.02	0.03	0.00	0.11	0.11	-0.00	0.02	0.02	-0.00
	(0.15)	(0.16)	(0.81)	(0.31)	(0.31)	(0.93)	(0.14)	(0.14)	(0.92)
rCARI using Income: Food Insecure	0.41	0.51	0.10**	0.64	0.73	0.09***	0.47	0.47	0.01
	(0.49)	(0.50)	(0.01)	(0.48)	(0.44)	(0.00)	(0.50)	(0.50)	(0.91)
Observations	610	232	842	760	656	1,416	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

A comparison of results using income and income change modules implemented over the phone (rCARI) and in F2F interviews is presented in Table 5. Aside from CAR, the results show that the module when implemented over the phone produces a significantly higher number of food insecurities compared to when asked in an F2F. This difference was between 9 and 10 percentage

points.

Next, we investigate how CARI aggregated using FES with income and income change variables modules collected in F2F surveys from the same households/respondents. This step allowed the comparison of these two indicators using the same respondents in the same modality. Results for the 3 pilot countries are presented in Table 6. The main conclusion is that CARI aggregated using income and income change variable significantly understates the numbers of food insecure (6% in Myanmar, 23% in Ecuador and 22% in CAR).

Table 6: Impact of Indicator Composition: Paired CARI-FES vs. CARI-Inc in F2F Surveys

	Myanmar			Ecuador			CAR		
Food Security Category	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff
Food Secure	0.08	0.16	0.08***	0.01	0.03	0.02	0.01	0.02	0.01*
	(0.01)	(0.01)	(0.01)	(0.00)	(0.01)	(0.01)	(0.00)	(0.01)	(0.01)
Marginally Food Secure	0.44	0.43	-0.01	0.23	0.33	0.10***	0.31	0.51	-0.20***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Moderately Food Insecure	0.44	0.39	-0.05**	0.59	0.53	-0.06**	0.60	0.45	-0.16***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Severely Food Insecure	0.03	0.02	-0.01	0.16	0.11	-0.06***	0.08	0.02	-0.06***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.01)	(0.02)
Food Insecure Group	0.47	0.41	-0.06***	0.76	0.64	-0.12***	0.68	0.47	-0.22***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Observations	610	610	610	760	760	760	294	294	294

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.
For CAR, the sample is limited to phone owners only.

Similar to the comparison between the FES and income variables in the F2F modality, we investigate the performance of these variables (FES and the income and income change variables) in the remote modality. Reported in Table 7, we find that the income and income change variable is conservatively under reporting the number of food insecure by between 12 to 24 percentage points (12 percent in Myanmar, 19 percent in Ecuador and 24 percent in CAR).

Table 7: Impact of Indicator Composition: Paired rCARI-FES vs. rCARI-Inc in Remote Surveys

	Myanmar			Ecuador			CAR		
Food Security Category	rCARI-FES	rCARI-Inc	Diff	rCARI-FES	rCARI-Inc	Diff	rCARI-FES	rCARI-Inc	Diff
Food Secure	0.03	0.07	0.05**	0.00	0.00	0.00	0.01	0.01	0.00
	(0.01)	(0.02)	(0.02)	(0.00)	(0.00)	(0.00)	(0.01)	(0.01)	(0.00)
Marginally Food Secure	0.34	0.41	0.07*	0.07	0.27	0.19***	0.29	0.52	0.24***
	(0.03)	(0.03)	(0.03)	(0.01)	(0.02)	(0.02)	(0.04)	(0.04)	(0.04)
Moderately Food Insecure	0.57	0.49	-0.09**	0.69	0.63	-0.07**	0.58	0.45	-0.13*
	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.04)	(0.04)	(0.05)
Severely Food Insecure	0.06	0.03	-0.03*	0.23	0.11	-0.13***	0.18	0.02	-0.11***
	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)	(0.01)	(0.03)	(0.01)	(0.03)
Food Insecure Group	0.63	0.51	-0.12***	0.92	0.73	-0.19***	0.71	0.47	-0.24***
	(0.03)	(0.03)	(0.03)	(0.01)	(0.02)	(0.02)	(0.04)	(0.04)	(0.04)
Observations	232	232	232	656	656	656	157	157	157

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.
For CAR, the sample is limited to phone owners only.

4.2 FCS

The results of the FCS analysis show that respondents tend to overestimate their consumption (report better food consumption) over the phone compared to when interviewed in F2F. A comparison of the FCS reported over the 2 modalities show statistical significance in Ecuador and CAR. However, we do not find statistical significance between the two in Myanmar. This is reflected in the 3-phase food consumption classification (acceptable, borderline and poor). The results show that more households in the F2F category significantly classified as having poor food consumption relative to households interviewed in the remote category (see Table 8 for details).

Table 8: Impact of Survey Modality on Food Consumption Score and Groups

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Consumption Score	44.32	43.75	-0.57	46.28	52.54	6.25***	39.76	47.63	7.87***
	(12.36)	(12.15)	(0.45)	(21.57)	(24.12)	(0.00)	(17.58)	(17.45)	(0.00)
FCS: Acceptable	0.54	0.48	-0.06	0.54	0.62	0.09***	0.40	0.58	0.18***
	(0.50)	(0.50)	(0.07)	(0.50)	(0.49)	(0.00)	(0.49)	(0.50)	(0.00)
FCS: Borderline	0.40	0.45	0.05	0.24	0.22	-0.02	0.35	0.29	-0.06
	(0.49)	(0.50)	(0.09)	(0.43)	(0.42)	(0.35)	(0.48)	(0.45)	(0.19)
FCS: Poor	0.06	0.07	0.00	0.22	0.16	-0.07**	0.25	0.13	-0.12**
	(0.24)	(0.25)	(0.79)	(0.42)	(0.36)	(0.00)	(0.43)	(0.34)	(0.00)
Consumption over the past 7 days (cereals and tubers)	7.00	7.00	0.00	5.38	4.93	-0.45***	5.87	6.24	0.36*
	(0.12)	(0.00)	(0.32)	(2.28)	(2.32)	(0.00)	(1.86)	(1.33)	(0.02)
Consumption over the past 7 days (pulses)	1.63	1.82	0.20*	2.00	2.50	0.50***	3.74	3.74	-0.01
	(1.46)	(1.55)	(0.03)	(2.06)	(2.24)	(0.00)	(2.33)	(2.08)	(0.98)
Consumption over the past 7 days (dairy products)	0.17	0.18	0.01	1.38	2.03	0.65***	0.50	1.14	0.64**
	(0.67)	(0.62)	(0.79)	(2.02)	(2.32)	(0.00)	(1.57)	(2.27)	(0.00)
Consumption over the past 7 days (protein-rich foods)	3.22	3.01	-0.21	3.03	3.48	0.44***	1.72	2.09	0.37*
	(2.05)	(1.97)	(0.09)	(2.42)	(2.39)	(0.00)	(1.81)	(1.81)	(0.04)
Consumption over the past 7 days (vegetables)	6.42	6.48	0.06	3.93	4.75	0.82***	3.73	5.64	1.92***
	(1.36)	(1.21)	(0.43)	(2.82)	(2.56)	(0.00)	(2.93)	(1.66)	(0.00)
Consumption over the past 7 days (fruit)	1.73	1.43	-0.30**	2.58	2.90	0.31*	0.54	0.72	0.18
	(1.92)	(1.67)	(0.01)	(2.57)	(2.61)	(0.02)	(1.31)	(1.34)	(0.17)
Consumption over the past 7 days (oil)	6.67	6.37	-0.30***	5.38	5.40	0.02	4.90	5.79	0.89***
	(1.21)	(1.31)	(0.00)	(2.52)	(2.34)	(0.87)	(2.37)	(1.81)	(0.00)
Consumption over the past 7 days (sugar)	0.82	0.85	0.03	5.39	5.64	0.25*	2.36	3.53	1.17***
	(1.50)	(1.58)	(0.74)	(2.57)	(2.25)	(0.05)	(2.60)	(2.78)	(0.00)
Observations	610	483	1,093	760	656	1,416	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

4.3 rCSI

A comparison of the reduced coping strategy adoption (rCSI) based on F2F and remote interviews is reported in Table 7. The results show that in Myanmar and CAR, households tend to report adopting more reduced coping strategies over the phone compared to F2F while, the opposite is found in Ecuador (where respondents are adopting significantly higher reduced coping strategies in the F2F).

Table 9: Impact of Survey Modality on Reduced Coping Strategy Index Classification

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Reduced Coping Strategies Index (rCSI)	5.43	9.24	3.81***	19.86	18.50	-1.36*	14.35	17.27	2.92**
	(7.46)	(8.87)	(0.00)	(12.41)	(11.68)	(0.03)	(9.97)	(8.88)	(0.00)
rCSI: Low	0.59	0.31	-0.29***	0.08	0.06	-0.02	0.06	0.04	-0.02
	(0.49)	(0.46)	(0.00)	(0.28)	(0.24)	(0.09)	(0.25)	(0.21)	(0.36)
rCSI: Median	0.33	0.53	0.20***	0.44	0.51	0.08**	0.70	0.57	-0.14**
	(0.47)	(0.50)	(0.00)	(0.50)	(0.50)	(0.00)	(0.46)	(0.50)	(0.00)
rCSI: Severe	0.08	0.17	0.09***	0.48	0.43	-0.05	0.23	0.39	0.16***
	(0.26)	(0.37)	(0.00)	(0.50)	(0.50)	(0.05)	(0.42)	(0.49)	(0.00)
Relied on less preferred, less expensive food	1.92	2.86	0.94***	4.41	4.56	0.15	3.92	2.75	-1.17***
	(2.20)	(2.26)	(0.00)	(2.58)	(2.71)	(0.29)	(2.00)	(1.84)	(0.00)
Borrowed food or relied on help from friends or relatives	0.59	0.56	-0.03	1.64	1.75	0.11	0.77	0.89	0.12
	(1.20)	(1.13)	(0.69)	(1.95)	(2.13)	(0.33)	(1.15)	(1.25)	(0.32)
Reduced portion size of meals at meals time	0.56	0.74	0.18*	3.80	3.27	-0.53***	1.98	2.65	0.67**
	(1.32)	(1.42)	(0.03)	(2.70)	(2.85)	(0.00)	(2.27)	(2.14)	(0.00)
Restricted consumption by adults in order for young-children to eat	0.52	1.23	0.72***	1.87	1.51	-0.36**	1.73	2.70	0.97***
	(1.39)	(1.78)	(0.00)	(2.38)	(2.24)	(0.00)	(1.90)	(1.83)	(0.00)
Reduced the number of meals eaten per day	0.22	0.82	0.60***	2.77	2.66	-0.11	1.74	2.00	0.26
	(0.90)	(1.67)	(0.00)	(2.52)	(2.70)	(0.43)	(2.38)	(2.12)	(0.24)
Observations	610	483	1,093	760	656	1,416	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

In terms of categorizing respondents into the 3 classes of rCSI, the remote surveys seem to classify more households in the severe category both in Myanmar and CAR than Ecuador.

4.4 LCS

The livelihood coping strategy adoption by respondents for none and stress level across countries follows the same trend (higher adoption in F2F than remote), however the results are mixed comparing the crisis and emergency levels. For example, while the result shows a significantly higher adoption of these categories by households taking the remote interviews in Myanmar and CAR, the opposite is observed for crisis level in Ecuador. Worthy of mention is the fact that in Ecuador, there is big jump in the adoption of emergency category strategies between F2F and remote (50% more households in the remote interview indicated adopted a strategy from this broad category).

Table 10: Impact of Survey Modality on Livelihood Coping Strategy Index

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Maximum Livelihood Coping: None	0.26	0.12	-0.14***	0.09	0.02	-0.07***	0.20	0.11	-0.09*
	(0.44)	(0.32)	(0.00)	(0.29)	(0.14)	(0.00)	(0.40)	(0.32)	(0.01)
Maximum Livelihood Coping: Stress level	0.46	0.41	-0.05	0.28	0.08	-0.20***	0.61	0.43	-0.19***
	(0.50)	(0.49)	(0.10)	(0.45)	(0.28)	(0.00)	(0.49)	(0.50)	(0.00)
Maximum Livelihood Coping: Crisis level	0.25	0.33	0.08**	0.42	0.19	-0.23***	0.14	0.26	0.13**
	(0.43)	(0.47)	(0.00)	(0.49)	(0.39)	(0.00)	(0.34)	(0.44)	(0.00)
Maximum Livelihood Coping: Emergency level	0.03	0.14	0.10***	0.21	0.70	0.50***	0.05	0.20	0.15***
	(0.18)	(0.35)	(0.00)	(0.41)	(0.46)	(0.00)	(0.22)	(0.40)	(0.00)
Observations	610	483	1,093	760	656	1,416	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

Table 11: Myanmar: Each Coping Strategy Used

	Myanmar		
Variable	F2F	Remote	Diff
Used Stress coping: Sold household assets/goods	0.07	0.13	0.06**
	(0.26)	(0.33)	(0.00)
Used Stress coping: Purchased food/non-food on credit (incur debts)	0.53	0.67	0.14***
	(0.50)	(0.47)	(0.00)
Used Stress coping: Spent savings	0.24	0.32	0.08**
	(0.42)	(0.47)	(0.00)
Used Stress coping: Borrowed money	0.42	0.65	0.23***
	(0.49)	(0.48)	(0.00)
Used Crisis coping: Sold productive assets or means of transport	0.02	0.06	0.04**
	(0.14)	(0.23)	(0.00)
Used Crisis coping: Reduced expenses on health	0.25	0.40	0.15***
	(0.43)	(0.49)	(0.00)
Used Crisis coping: Withdrew children from school	0.01	0.06	0.05***
	(0.10)	(0.25)	(0.00)
Used Emergency coping: Send children to work for household income	0.01	0.05	0.04***
	(0.11)	(0.23)	(0.00)
Used Emergency coping: Begged and/or scavenged	0.01	0.03	0.02*
	(0.11)	(0.18)	(0.02)
Used Emergency coping: Engaged in illegal income activities	0.02	0.07	0.05***
	(0.13)	(0.26)	(0.00)
Observations	610	483	1,093

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

Table 12: Ecaudor: Each Coping Strategy Used

Variable	Ecuador F2F	Remote	Diff
Used Stress coping: Sold household assets/goods	0.10	0.15	0.05**
	(0.31)	(0.36)	(0.01)
Used Stress coping: Purchased food/non-food on credit (incur debts)	0.39	0.31	-0.08**
	(0.49)	(0.46)	(0.00)
Used Stress coping: Spent savings	0.62	0.65	0.03
	(0.49)	(0.48)	(0.18)
Used Stress coping: Borrowed money	0.64	0.64	0.01
	(0.48)	(0.48)	(0.80)
Used Crisis coping: Sold productive assets or means of transport	0.05	0.08	0.03
	(0.23)	(0.27)	(0.06)
Used Crisis coping: Reduced expenses on health and education	0.57	0.61	0.04
	(0.50)	(0.49)	(0.10)
Used Crisis coping: Withdrew children from school	0.08	0.11	0.03
	(0.27)	(0.31)	(0.07)
Used Emergency coping: Send children to work for household income	0.08	0.09	0.01
	(0.27)	(0.28)	(0.53)
Used Emergency coping: Begged and/or scavenged	0.10	0.23	0.13***
	(0.29)	(0.42)	(0.00)
Used Emergency coping: Engaged in illegal income activities	0.09	0.58	0.49***
	(0.28)	(0.49)	(0.00)
Observations	760	656	1,416

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

Table 13: CAR: Each Coping Strategy Used

Variable	CAR F2F	Remote	Diff
Used Stress coping: Sold household assets/goods	0.28	0.46	0.18***
	(0.45)	(0.50)	(0.00)
Used Stress coping: Reduced expenses on health and education	0.23	0.43	0.20***
	(0.42)	(0.50)	(0.00)
Used Stress coping: Spent savings	0.47	0.50	0.03
	(0.50)	(0.50)	(0.50)
Used Stress coping: Borrowed money	0.46	0.53	0.07
	(0.50)	(0.50)	(0.18)
Used Crisis coping: Sold productive assets or means of transport	0.07	0.26	0.19***
	(0.26)	(0.44)	(0.00)
Used Crisis coping: Moved to somewhere else	0.05	0.14	0.09**
	(0.23)	(0.35)	(0.01)
Used Crisis coping: Withdrew children from school	0.03	0.05	0.02
	(0.17)	(0.22)	(0.32)
Used Emergency coping: Sold houses	0.03	0.14	0.11***
	(0.17)	(0.35)	(0.00)
Used Emergency coping: Begged and/or scavenged	0.01	0.07	0.06**
	(0.12)	(0.26)	(0.01)
Used Emergency coping: Engaged in illegal income activities	0.01	0.03	0.02
	(0.10)	(0.16)	(0.27)
Observations	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

4.5 FES

The aggregated FES show that across the 3 pilot countries, respondents over the phone tend to state their FES share on average by between 5% and 8% compared to F2F. This is significantly reflected in the 75% and above category.

Table 14: Impact of Survey Modality on Expenditure and Food Expenditure Share

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Expenditure Share	0.66	0.71	0.05***	0.71	0.79	0.08***	0.67	0.73	0.06*
	(0.15)	(0.14)	(0.00)	(0.19)	(0.16)	(0.00)	(0.20)	(0.24)	(0.01)
FES < 50 percent	0.13	0.08	-0.06*	0.15	0.05	-0.10***	0.20	0.16	-0.04
	(0.34)	(0.27)	(0.01)	(0.36)	(0.22)	(0.00)	(0.40)	(0.37)	(0.27)
FES: 50-65 percent	0.30	0.22	-0.08*	0.17	0.09	-0.08***	0.23	0.17	-0.06
	(0.46)	(0.41)	(0.01)	(0.38)	(0.29)	(0.00)	(0.42)	(0.38)	(0.15)
FES: 65-75 percent	0.27	0.26	-0.01	0.19	0.18	-0.01	0.13	0.13	-0.01
	(0.44)	(0.44)	(0.76)	(0.39)	(0.38)	(0.54)	(0.34)	(0.33)	(0.87)
FES: >= 75 percent	0.30	0.45	0.15***	0.49	0.68	0.19***	0.44	0.54	0.10*
	(0.46)	(0.50)	(0.00)	(0.50)	(0.47)	(0.00)	(0.50)	(0.50)	(0.04)
Observations	610	232	842	760	656	1,416	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.
In Myanmar, the Short Form in Remote Survey was dropped due to translation issues.

4.6 Income

The income variable captures the respondent/household income source and income change experienced in the last 12 months prior to the interview. The results show that more respondents/households reported more income sources, with stability than when asked in the F2F. The result remains consistent in all 3 pilot countries.

Table 15: Impact of Survey Modality on Income

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Income: Regular and Unchanged/Increasing	0.22	0.19	-0.03	0.19	0.25	0.06**	0.45	0.26	-0.19***
	(0.41)	(0.39)	(0.30)	(0.40)	(0.44)	(0.01)	(0.50)	(0.44)	(0.00)
Income: Regular and Reduced OR Irregular and Unchanged/Increasing	0.39	0.41	0.03	0.38	0.41	0.03	0.42	0.50	0.08
	(0.49)	(0.49)	(0.36)	(0.49)	(0.49)	(0.25)	(0.49)	(0.50)	(0.13)
Income: Irregular and Reduced	0.28	0.33	0.04	0.26	0.24	-0.01	0.12	0.23	0.11**
	(0.45)	(0.47)	(0.11)	(0.44)	(0.43)	(0.54)	(0.32)	(0.42)	(0.00)
Income: No income sources	0.11	0.07	-0.05**	0.17	0.09	-0.08***	0.01	0.01	-0.00
	(0.32)	(0.25)	(0.01)	(0.37)	(0.29)	(0.00)	(0.12)	(0.11)	(0.94)
Observations	610	483	1,093	760	656	1,416	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

In the next section, we will explore further the individual indicators used as inputs for calculating CARI. The indicators are compared across modalities (F2F and Remote).

5 Impact of Form Length on FCS and Expenditure/FES

As a secondary objective, the study also tested a modified version of FCS and FES modules. We test an aggregated version (shortened version) against the standard module (long version). We also tested the performance of LCS-EN against LCS-FS modules. This would allow recommendation for the use of the shorter version of the FES module over the phone or in F2F surveys to reduce interviewee fatigue. The results of this section of the study (comparison of short and long (standard model length).

5.1 FCS Full Subgroup vs. FCS Short

In terms of FCS (see Table 16), the results are not significantly different between short and long modules in Myanmar and Ecuador except for CAR where the short version significantly overstates food consumption by 4.35 points.

Table 16: Impact of Module Length on FCS Reporting

	Myanmar F2F			Ecuador F2F			CAR F2F		
Variable	Long	Short	Diff	Long	Short	Diff	Long	Short	Diff
Food Consumption Score	44.67	43.96	-0.71	46.42	46.16	-0.26	37.57	41.92	4.35*
	(12.47)	(12.27)	(0.48)	(21.61)	(21.55)	(0.87)	(16.98)	(17.95)	(0.03)
FCS: Acceptable	0.54	0.53	-0.02	0.54	0.53	-0.02	0.32	0.48	0.16**
	(0.50)	(0.50)	(0.69)	(0.50)	(0.50)	(0.65)	(0.47)	(0.50)	(0.01)
FCS: Borderline	0.38	0.42	0.04	0.23	0.26	0.03	0.42	0.28	-0.14*
	(0.49)	(0.49)	(0.28)	(0.42)	(0.44)	(0.29)	(0.49)	(0.45)	(0.01)
FCS: Poor	0.08	0.05	-0.03	0.23	0.21	-0.02	0.26	0.24	-0.02
	(0.26)	(0.22)	(0.18)	(0.42)	(0.41)	(0.58)	(0.44)	(0.43)	(0.74)
Consumption over the past 7 days (cereals and tubers)	6.99	7.00	0.01	5.38	5.38	-0.00	5.79	5.95	0.17
	(0.17)	(0.00)	(0.32)	(2.31)	(2.25)	(1.00)	(1.96)	(1.75)	(0.45)
Consumption over the past 7 days (pulses)	1.60	1.66	0.06	1.91	2.09	0.18	3.71	3.78	0.08
	(1.42)	(1.50)	(0.62)	(2.07)	(2.05)	(0.23)	(2.21)	(2.45)	(0.77)
Consumption over the past 7 days (dairy products)	0.16	0.17	0.01	1.35	1.40	0.06	0.55	0.45	-0.10
	(0.77)	(0.56)	(0.86)	(1.96)	(2.08)	(0.70)	(1.61)	(1.53)	(0.60)
Consumption over the past 7 days (protein-rich foods)	3.35	3.10	-0.25	3.06	3.00	-0.06	1.66	1.78	0.11
	(2.04)	(2.04)	(0.13)	(2.45)	(2.39)	(0.73)	(1.69)	(1.92)	(0.59)
Consumption over the past 7 days (vegetables)	6.37	6.47	0.10	4.00	3.87	-0.13	2.10	5.34	3.24***
	(1.40)	(1.32)	(0.36)	(2.79)	(2.86)	(0.53)	(2.76)	(2.08)	(0.00)
Consumption over the past 7 days (fruit)	1.76	1.69	-0.07	2.77	2.40	-0.38*	0.46	0.62	0.16
	(1.97)	(1.88)	(0.64)	(2.59)	(2.53)	(0.04)	(1.26)	(1.36)	(0.29)
Consumption over the past 7 days (oil)	6.66	6.68	0.02	5.59	5.17	-0.42*	4.69	5.10	0.41
	(1.21)	(1.21)	(0.84)	(2.42)	(2.59)	(0.02)	(2.39)	(2.34)	(0.14)
Consumption over the past 7 days (sugar)	0.81	0.83	0.03	5.45	5.32	-0.14	2.25	2.47	0.22
	(1.58)	(1.41)	(0.83)	(2.55)	(2.59)	(0.47)	(2.39)	(2.79)	(0.47)
Observations	305	305	610	377	383	760	146	148	294

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

5.2 Expenditure Long vs. Short

The FES however remains stable showing no statistical significance between the short and long modules in all pilot countries (see Tables 17). The results are similar for the two versions used in the remote surveys (showing no significant difference between short and long).

Table 17: Impact of Module Length on Expenditure and FES in F2F Surveys

	Myanmar F2F			Ecuador F2F			CAR F2F		
Variable	Long	Short	Diff	Long	Short	Diff	Long	Short	Diff
Food Expenditure Share	0.66	0.66	0.00	0.72	0.70	-0.02	0.67	0.67	0.00
	(0.15)	(0.14)	(0.89)	(0.19)	(0.18)	(0.14)	(0.19)	(0.20)	(0.92)
FES < 50 percent	0.15	0.12	-0.03	0.14	0.16	0.03	0.20	0.20	0.00
	(0.36)	(0.32)	(0.24)	(0.34)	(0.37)	(0.30)	(0.40)	(0.40)	(0.93)
FES: 50-65 percent	0.29	0.31	0.03	0.18	0.17	-0.00	0.23	0.22	-0.01
	(0.45)	(0.46)	(0.48)	(0.38)	(0.38)	(1.00)	(0.42)	(0.42)	(0.84)
FES: 65-75 percent	0.26	0.28	0.02	0.17	0.21	0.04	0.15	0.11	-0.04
	(0.44)	(0.45)	(0.58)	(0.38)	(0.41)	(0.17)	(0.36)	(0.32)	(0.37)
FES: >= 75 percent	0.30	0.29	-0.01	0.52	0.45	-0.07	0.42	0.46	0.04
	(0.46)	(0.46)	(0.72)	(0.50)	(0.50)	(0.07)	(0.49)	(0.50)	(0.47)
Observations	305	305	610	377	383	760	146	148	294

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

Table 18: Impact of Module Length on Expenditure and FES in Remote Surveys

	Ecuador RM			CAR RM		
Variable	Long	Short	Diff	Long	Short	Diff
Food Expenditure Share	0.78	0.80	0.02	0.65	0.78	0.12**
	(0.15)	(0.17)	(0.06)	(0.26)	(0.21)	(0.00)
FES < 50 percent	0.04	0.06	0.02	0.22	0.12	-0.10
	(0.20)	(0.24)	(0.33)	(0.41)	(0.33)	(0.12)
FES: 50-65 percent	0.08	0.10	0.02	0.18	0.16	-0.02
	(0.28)	(0.30)	(0.40)	(0.39)	(0.37)	(0.73)
FES: 65-75 percent	0.23	0.13	-0.10***	0.18	0.09	-0.10
	(0.42)	(0.33)	(0.00)	(0.39)	(0.28)	(0.09)
FES: >= 75 percent	0.65	0.71	0.07	0.42	0.63	0.22**
	(0.48)	(0.45)	(0.07)	(0.50)	(0.49)	(0.01)
Observations	324	332	656	65	92	157

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.
Myanmar is excluded due to translation issue.

Table 19: Impact of Module Length on Expenditure and FES: F2F vs. Remote Long

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote Long	Diff	F2F	Remote Long	Diff	F2F	Remote Long	Diff
Food Expenditure Share	0.66	0.71	-0.02	0.71	0.78	-0.05***	0.67	0.65	0.01
	(0.15)	(0.14)	(0.07)	(0.19)	(0.15)	(0.00)	(0.20)	(0.26)	(0.77)
FES < 50 percent	0.13	0.08	-0.00	0.15	0.04	0.07**	0.20	0.22	-0.00
	(0.34)	(0.27)	(0.97)	(0.36)	(0.20)	(0.00)	(0.40)	(0.41)	(0.98)
FES: 50-65 percent	0.30	0.21	0.06	0.17	0.08	0.04	0.23	0.18	0.00
	(0.46)	(0.41)	(0.08)	(0.38)	(0.28)	(0.07)	(0.42)	(0.39)	(0.91)
FES: 65-75 percent	0.27	0.26	0.02	0.19	0.23	0.01	0.13	0.18	-0.05
	(0.44)	(0.44)	(0.56)	(0.39)	(0.42)	(0.64)	(0.34)	(0.39)	(0.21)
FES: >= 75 percent	0.30	0.45	-0.07*	0.49	0.65	-0.12***	0.44	0.42	0.04
	(0.46)	(0.50)	(0.03)	(0.50)	(0.48)	(0.00)	(0.50)	(0.50)	(0.43)
Observations	610	233	843	760	324	1,084	294	65	359

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

Table 20: Impact of Module Length on Expenditure and FES: F2F vs. Remote Short

	Ecuador			CAR		
Variable	F2F	Remote Short	Diff	F2F	Remote Short	Diff
Food Expenditure Share	0.71	0.80	-0.05***	0.67	0.78	0.01
	(0.19)	(0.17)	(0.00)	(0.20)	(0.21)	(0.77)
FES < 50 percent	0.15	0.06	0.07**	0.20	0.12	-0.00
	(0.36)	(0.24)	(0.00)	(0.40)	(0.33)	(0.98)
FES: 50-65 percent	0.17	0.10	0.04	0.23	0.16	0.00
	(0.38)	(0.30)	(0.07)	(0.42)	(0.37)	(0.91)
FES: 65-75 percent	0.19	0.13	0.01	0.13	0.09	-0.05
	(0.39)	(0.33)	(0.64)	(0.34)	(0.28)	(0.21)
FES: >= 75 percent	0.49	0.71	-0.12***	0.44	0.63	0.04
	(0.50)	(0.45)	(0.00)	(0.50)	(0.49)	(0.43)
Observations	760	332	1,084	294	92	359

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.
Myanmar is excluded due to translation issue.

6 Impact of LCS for Essential Needs or Food Security

6.1 Overall

Table 21: Myanmar: LCS for Essential Needs vs. Food Security

	Myanmar F2F			Ecuador F2F			CAR F2F		
Variable	EN	FS	Diff	EN	FS	Diff	EN	FS	Diff
Maximum Livelihood Coping: None	0.25	0.26	0.01	0.11	0.07	-0.04	0.19	0.21	0.02
	(0.44)	(0.44)	(0.78)	(0.32)	(0.26)	(0.05)	(0.40)	(0.41)	(0.71)
Maximum Livelihood Coping: Stress level	0.47	0.45	-0.02	0.26	0.30	0.05	0.60	0.62	0.02
	(0.50)	(0.50)	(0.63)	(0.44)	(0.46)	(0.16)	(0.49)	(0.49)	(0.74)
Maximum Livelihood Coping: Crisis level	0.26	0.24	-0.01	0.45	0.40	-0.05	0.14	0.13	-0.02
	(0.44)	(0.43)	(0.71)	(0.50)	(0.49)	(0.15)	(0.35)	(0.34)	(0.70)
Maximum Livelihood Coping: Emergency level	0.02	0.05	0.02	0.18	0.23	0.05	0.06	0.04	-0.02
	(0.15)	(0.21)	(0.12)	(0.39)	(0.42)	(0.11)	(0.24)	(0.20)	(0.41)
Observations	305	305	610	377	383	760	146	148	294

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

6.2 Specific Strategy

Table 22: Myanmar: LCS for Essential Needs vs. Food Security

	Myanmar F2F		
Variable	LCS EN	LCS FS	Diff
Used Stress coping: Sold household assets/goods	0.07	0.08	0.01
	(0.25)	(0.27)	(0.62)
Used Stress coping: Purchased food/non-food on credit (incur debts)	0.54	0.53	-0.01
	(0.50)	(0.50)	(0.81)
Used Stress coping: Spent savings	0.22	0.25	0.02
	(0.42)	(0.43)	(0.48)
Used Stress coping: Borrowed money	0.46	0.38	-0.08*
	(0.50)	(0.49)	(0.04)
Used Crisis coping: Sold productive assets or means of transport	0.02	0.02	-0.00
	(0.15)	(0.14)	(0.79)
Used Crisis coping: Reduced expenses on health	0.26	0.25	-0.01
	(0.44)	(0.43)	(0.82)
Used Crisis coping: Withdrew children from school	0.01	0.01	0.00
	(0.10)	(0.10)	(0.99)
Used Emergency coping: Send children to work for household income	0.00	0.02	0.02
	(0.06)	(0.14)	(0.06)
Used Emergency coping: Begged and/or scavenged	0.01	0.01	-0.00
	(0.11)	(0.10)	(0.71)
Used Emergency coping: Engaged in illegal income activities	0.01	0.02	0.01
	(0.10)	(0.15)	(0.20)
Observations	305	305	610

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

Table 23: Ecuador: LCS for Essential Needs vs. Food Security

	Ecuador F2F		
Variable	LCS EN	LCS FS	Diff
Used Stress coping: Sold household assets/goods	0.10	0.11	0.01
	(0.30)	(0.31)	(0.60)
Used Stress coping: Purchased food/non-food on credit (incur debts)	0.37	0.41	0.04
	(0.48)	(0.49)	(0.22)
Used Stress coping: Spent savings	0.61	0.63	0.01
	(0.49)	(0.48)	(0.69)
Used Stress coping: Borrowed money	0.62	0.66	0.04
	(0.49)	(0.48)	(0.22)
Used Crisis coping: Sold productive assets or means of transport	0.06	0.05	-0.01
	(0.23)	(0.22)	(0.59)
Used Crisis coping: Reduced expenses on health and education	0.55	0.58	0.04
	(0.50)	(0.49)	(0.32)
Used Crisis coping: Withdrew children from school	0.09	0.07	-0.02
	(0.29)	(0.25)	(0.20)
Used Emergency coping: Send children to work for household income	0.07	0.08	0.01
	(0.26)	(0.27)	(0.73)
Used Emergency coping: Begged and/or scavenged	0.07	0.12	0.05*
	(0.25)	(0.33)	(0.01)
Used Emergency coping: Engaged in illegal income activities	0.08	0.09	0.00
	(0.28)	(0.28)	(0.95)
Observations	377	383	760

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

Table 24: CAR: LCS for Essential Needs vs. Food Security

	CAR F2F		
Variable	LCS EN	LCS FS	Diff
Used Stress coping: Sold household assets/goods	0.32	0.24	-0.08
	(0.47)	(0.43)	(0.14)
Used Stress coping: Reduced expenses on health and education	0.24	0.22	-0.02
	(0.43)	(0.41)	(0.63)
Used Stress coping: Spent savings	0.45	0.49	0.05
	(0.50)	(0.50)	(0.41)
Used Stress coping: Borrowed money	0.51	0.42	-0.09
	(0.50)	(0.50)	(0.13)
Used Crisis coping: Sold productive assets or means of transport	0.09	0.06	-0.03
	(0.29)	(0.24)	(0.36)
Used Crisis coping: Moved to somewhere else	0.07	0.04	-0.03
	(0.25)	(0.20)	(0.29)
Used Crisis coping: Withdrew children from school	0.03	0.03	0.01
	(0.16)	(0.18)	(0.75)
Used Emergency coping: Sold houses	0.05	0.01	-0.03
	(0.21)	(0.12)	(0.09)
Used Emergency coping: Begged and/or scavenged	0.01	0.02	0.01
	(0.08)	(0.14)	(0.32)
Used Emergency coping: Engaged in illegal income activities	0.01	0.01	0.01
	(0.08)	(0.12)	(0.57)
Observations	146	148	294

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

7 Challenges of the Study by Country

In this section, the main challenges faced in the 3 pilot countries are listed.

7.1 Myanmar

1. Outdated listing data 2. Poor/unstable network coverage 3. Mobility of respondents/households 4. Absence of baseline data of the sample 5. Limited coverage of the study population (limited to IDP camps) 6. Typhoon season affected the implementation period. 7. Restricted access to the field 8. Distrust from respondents

7.2 Ecuador

1. Issues of security for enumerators in some communities. a. Some parts of the country were not accessible due to safety concerns for the face-to-face enumerators. 2. Misunderstanding with the local authorities preventing field access by F2F team. a. Dues to political reasons, some local authorities refused enumerators access leading to dropping of these communities. b. Distrust from local authorities and in some cases respondents. 3. Invalid respondent of household contact information (collected during the listing phase) 4. Poor phone coverage in remote areas 5. Disproportionate ownership of phones between urban and rural areas (low ownership in rural areas) 6. Low response rate particularly among the remote sample a. High attrition between the time of receiving consent and actual interview date.

7.3 CAR

1. Restricted access to the field because of deteriorating security situation 2. Bad phone network/infrastructure a. Most parts of the country do not have phone network coverage. b. In places where there is coverage, poor infrastructure and constant electricity cuts affected call quality. c. No access to electricity for charging phones particularly in rural areas. 3. Low phone ownership in the population a. Only 25% of households had phones (similarly to the national statistic) 4. Logistical challenges (Enumerator deployment) a. Depended on the UNHAS to deploy enumerators this affected how quickly deployment could be done (flights to study areas twice a week) b. Acquisition and distribution of phones/SIM cards-Challenges with moving them to the field for distribution. 5. Low response rate in the remote arm (High Attrition) 6. Lack of remote data collection platform by the Vendor (ICASEES) 7. Weather/rainy season

8 Summary and conclusions

Table 25: Impact of Survey Modality on Food Consumption Score and Groups

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Consumption Score	44.32	43.75	-0.57	46.28	52.54	6.25***	39.76	47.63	7.87***
	(12.36)	(12.15)	(0.45)	(21.57)	(24.12)	(0.00)	(17.58)	(17.45)	(0.00)
FCS: Acceptable	0.54	0.48	-0.06	0.54	0.62	0.09***	0.40	0.58	0.18***
	(0.50)	(0.50)	(0.07)	(0.50)	(0.49)	(0.00)	(0.49)	(0.50)	(0.00)
FCS: Borderline	0.40	0.45	0.05	0.24	0.22	-0.02	0.35	0.29	-0.06
	(0.49)	(0.50)	(0.09)	(0.43)	(0.42)	(0.35)	(0.48)	(0.45)	(0.19)
FCS: Poor	0.06	0.07	0.00	0.22	0.16	-0.07**	0.25	0.13	-0.12**
	(0.24)	(0.25)	(0.79)	(0.42)	(0.36)	(0.00)	(0.43)	(0.34)	(0.00)
Observations	610	483	1,093	760	656	1,416	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

Table 26: Myanmar Sample Balance Table

	Myanmar		
Variable	F2F	Remote	Diff
Respondent Age	42.24	38.26	-3.98***
	(13.41)	(13.58)	(0.00)
Female Respondent	0.81	0.80	-0.01
	(0.39)	(0.40)	(0.61)
Head of Household Respondent	0.42	0.35	-0.07*
	(0.49)	(0.48)	(0.02)
Household Size	5.52	5.49	-0.03
	(2.20)	(2.18)	(0.84)
Number of household members contributing to income	1.41	1.41	-0.00
	(0.85)	(0.82)	(0.98)
Number of household income sources	1.18	1.24	0.07
	(0.68)	(0.59)	(0.09)
Observations	610	483	1,093

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.

Table 27: Ecuador Sample Balance Table

Variable	F2F	Ecuador Remote	Diff
Respondent Age	46.19	43.00	-3.19**
	(17.97)	(16.84)	(0.00)
Female Respondent	0.77	0.76	-0.01
	(0.42)	(0.43)	(0.51)
Head of Household Respondent	0.60	0.53	-0.07*
	(0.49)	(0.50)	(0.01)
Household Size	4.37	4.57	0.19
	(2.24)	(2.35)	(0.11)
Number of household members contributing to income	1.47	1.45	-0.01
	(0.84)	(0.80)	(0.75)
Number of household income sources	1.40	1.50	0.11**
	(0.72)	(0.80)	(0.01)
Observations	760	656	1,416

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.
Figures might be changed slightly after roster inconsistency corrections.

Table 28: CAR Sample Balance Table

Variable	F2F	CAR Remote	Diff
List respondent age	36.27	39.03	2.76*
	(12.55)	(12.38)	(0.03)
Household head is the respondent	0.72	0.78	0.05
	(0.45)	(0.42)	(0.21)
Female Household Head	0.19	0.25	0.06
	(0.39)	(0.44)	(0.12)
Married Household Head	0.27	0.27	0.01
	(0.44)	(0.45)	(0.85)
Household head years of education	7.55	7.92	0.38
	(4.13)	(4.69)	(0.40)
Displaced household	0.09	0.15	0.06
	(0.29)	(0.36)	(0.07)
Household size	7.72	8.39	0.67*
	(3.44)	(3.40)	(0.05)
Number of household member younger than 15	4.04	4.31	0.27
	(2.27)	(2.03)	(0.20)
Number of household member older than 60	0.21	0.21	-0.00
	(0.49)	(0.44)	(0.99)
Has household member with disability/chronic disease	0.30	0.31	0.01
	(0.46)	(0.46)	(0.83)
Number of rooms	2.89	3.13	0.24
	(1.24)	(1.27)	(0.05)
Household Crowding Index	3.02	2.97	-0.05
	(1.70)	(1.26)	(0.73)
Household has impoverished roof	0.68	0.63	-0.05
	(0.47)	(0.48)	(0.33)
Household has impoverished floor	0.90	0.92	0.02
	(0.30)	(0.28)	(0.50)
Observations	294	157	451

Note: Significance: * p<0.05, ** p<0.01, *** p<0.001.
Listing answers were used in this table.